MATHEMATICAL GAMES

How rectangles, including squares, can be divided into squares of unequal size

by Martin Gardner

Can a square be subdivided into smaller squares of which no two are alike? This enormously difficult problem was long thought to be unsolvable, but now it has been defeated by translating it into electrical-network theory and back into plane geometry. Here William T. Tutte, assistant professor of mathematics at the University of Toronto, presents a fascinating account of how he and three fellow students at the University of Cambridge finally squared the square.

"This is the story of a mathematical research conducted by four students of Trinity College, Cambridge, in the years 1936-38. One was the author of this article. Another was C. A. B. Smith, now a statistical geneticist at University College London. He is also well known as a writer on the theory of games and the counterfeit-coin problem. Another was Arthur H. Stone, now researching recondite regions of point-set topology. He is one of the inventors of the flexagons described in an earlier issue of *Scientific American*. The fourth was R. L. Brooks. He has now left the academic world for the British Civil Service. But he retains an enthusiasm for mathematical recreations, and an important theorem in the theory of graph colorings bears his name. These four students referred to themselves with characteristic modesty as the 'Important Members' of the Trinity Mathematical Society.

"In 1936 there were a few references in the literature on the problem of cutting up a rectangle into squares of unequal size. Thus it was known that a rectangle of which one side was 32 units long and the other 33 units could be dissected into nine squares with sides of 1, 4, 7, 8, 9, 10, 14, 15 and 18 units (*Figure 1*). Stone was intrigued by a statement in Henry Ernest Dudeney's *The Canterbury Puzzles* which seemed to imply that it was impossible to cut up a *square* into unequal smaller squares. He tried to prove the impossibility, without success. He did, however, discover a dissection of the rectangle of sides 176 and 177 into 11 squares.

"This partial success fired the imagination of Stone and his three friends, and soon they were spending much time on the problem. The construction of 'perfect rectangles' (the term used by the group to denote any rectangle cut up into unequal squares) proved to be quite easy. The method used was as follows. First we sketch a rectangle cut up into rectangles, as in Figure 2. We then think of the diagram as a bad drawing of a squared rectangle in which the small rectangles are really squares, and we work out by elementary algebra what the relative sizes of the squares must be on this assumption. Thus in Figure 2 we denote the sides of two adjacent small squares by x and y. We can say that the side of the square immediately below them is x plus y, and then that the side of the square next on the left is x plus $2y$, and so on. Proceeding in this way we get the formulas shown in Figure 2 for the sides of the 11 small squares. These formulas make the squares fit together exactly except along the one segment AB. But we can make them fit on AB too by choosing x and y to satisfy the equation: $(3x + y) + (3x - 3y) = (14y - 3x)$. This reduces to $16y = 9x$. Accordingly we write down $x = 16$ and $y = 9$. This gives the perfect rectangle of Figure 3, which is the one first found by Stone.

"Sometimes this method gave negative values for the sides of small squares. It was found, however, that such negative squares could always be converted into positive ones by minor modifications of the original diagram. In some of the more complicated diagrams it proved necessary to start with three unknown squares of sides x, y and z, and to solve two linear equations instead of one. Sometimes the squared rectangles obtained proved not to be perfect, and the attempt was considered a failure. Fortunately this did not happen very often. We recorded only 'simple' perfect rectangles, that is, perfect rectangles containing no smaller perfect rectangles.

"In this first stage of the research large numbers of perfect rectangles were constructed, their 'order' or number of component squares ranging from nine to 26. In the final form of each rectangle the sides of the squares were represented as integers without a common factor. Of course we all hoped that if we constructed enough perfect rectangles by this method we would eventually obtain one which was a perfect square. But as the list of perfect rectangles lengthened, this hope faded.

"In the next stage of the research we abandoned experiment for theory. We tried to represent squared rectangles by diagrams of different kinds. The last of these diagrams, introduced by Smith, suddenly made our problem part of the theory of electrical networks.

"Figure 4 shows a perfect rectangle together with its Smith diagram. Each horizontal line in the rectangle is represented by a dot, or 'terminal,' in the diagram. In the figure the terminal is placed so that it lies on a continuation to the right of its corresponding horizontal line. Any square in the rectangle is bounded above and below by two of the horizontal lines. Accordingly it is represented by a line, or 'wire,' that joins the two corresponding terminals. We imagine that an electric current can flow in each wire. The magnitude of the current is numerically equal to the side of the corresponding square, and its direction is from the terminal representing the upper horizontal line to the terminal representing the lower one.

"Surprisingly enough, the electric currents assigned by the above rule do obey Kirchhoff's laws for the flow of current in a network, provided we take each wire to be of unit resistance. Kirchhoff's first law states that, except at the poles, the algebraic sum of the currents flowing to any terminal is zero. This corresponds to the fact that the sum of the sides of the squares bounded below by a given horizontal line is equal to the sum of the sides of the squares bounded above the same line, provided of course that the line is not one of the horizontal sides of the rectangle. The second law says that the algebraic sum of the currents in any circuit is zero. This is equivalent to saying that when we describe the circuit, the net corresponding change of level in the rectangle must be zero.

"The total current entering the network at the positive pole, or leaving it at the negative pole, is evidently equal to the horizontal side of the rectangle, and

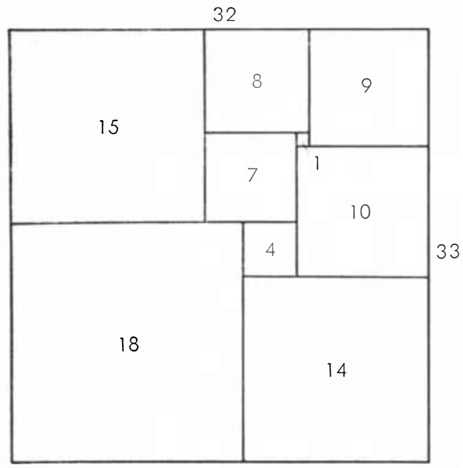

Figure 1

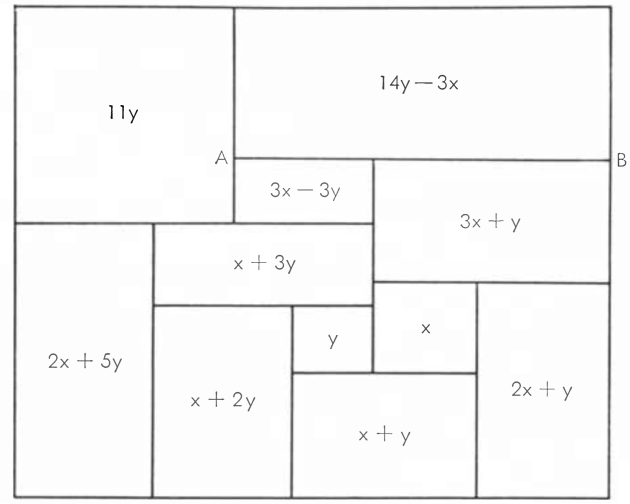

Figure 2

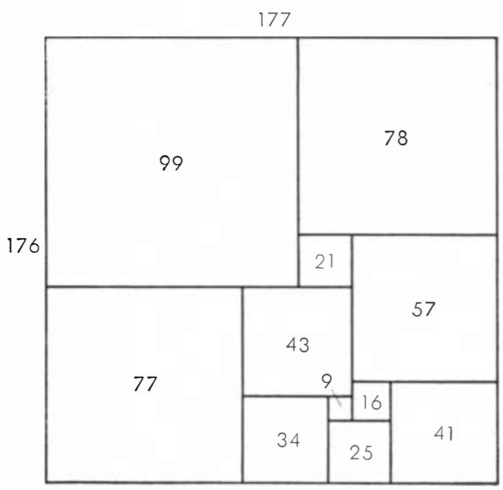

Figure 3

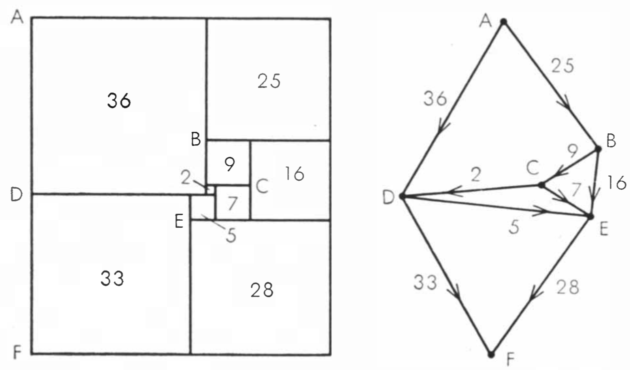

Figure 4

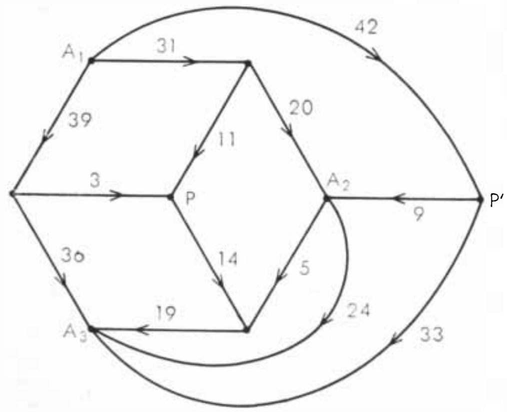

Figure 5

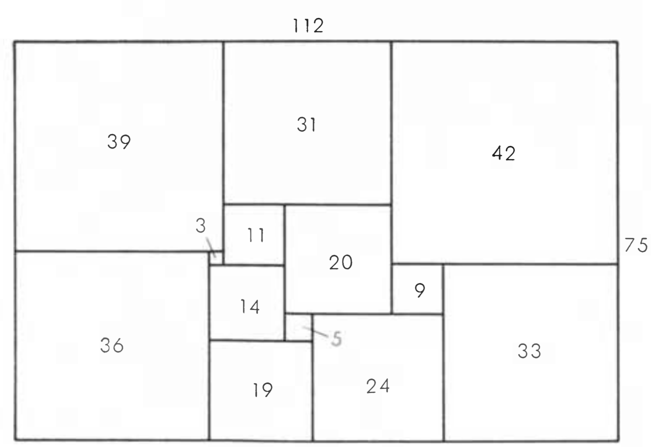

Figure 6

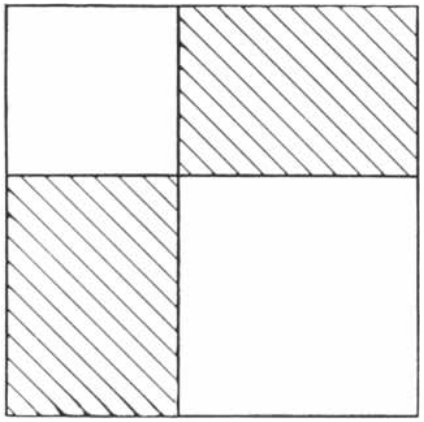

Figure 7

the potential difference between the two poles is equal to the vertical side.

"The discovery of this electrical analogy was important to us because it linked our problem with an established theory. We could now borrow from the theory of electrical networks and obtain formulas for the currents in a general Smith diagram and the sizes of the corresponding components squares. The main results of this borrowing can be summarized as follows. With each electrical network there is associated a number calculated from the structure of the network, without any reference to which particular pair of terminals is chosen as poles. We called this number the complexity of the network. If the units of measurement for the corresponding rectangle are chosen so that the horizontal side is equal to the complexity, then the sides of the component squares are all whole numbers. Moreover, the vertical side is equal to the complexity of another network obtained from the first by identifying the two poles.

"The numbers giving the side of the rectangle and its component squares in this system of measurement were respectively called the 'full' sides and the 'full' elements of the rectangle. For some rectangles the full elements have a common factor greater than unity. In such cases, dividing by the common factor gives the 'reduced' sides and elements. It was the reduced sides and elements that had been recorded in our catalog.

"The discovery of the Smith diagram simplified the procedure for producing and classifying simple squared rectangles. It was an easy matter to list all the permissible electrical networks up to 11 wires, and to calculate all the corresponding squared rectangles. We then found that there were no perfect rectangles below the ninth order, and only two of the ninth (*Figures 1 and 4*). There were six simple ones of the 10th order and 22 of the 11th. The catalog then advanced, though more slowly, through the 12th order (67 simple perfect rectangles) and into the 13th.

"It was a pleasing recreation to work out perfect rectangles corresponding to networks with a high degree of symmetry. We considered, for example, the network defined by a cube, with corners for terminals and edges for wires. This failed to yield any perfect rectangles. However, when complicated by a diagonal wire across one face, and flattened into a plane, it gave the Smith diagram of Figure 5 and the corresponding perfect rectangle of Figure 6. This rectangle is especially interesting because its reduced elements are unusually small for the 13th order. The common factor of the full elements is six. Brooks was so pleased with the rectangle that he made a jigsaw puzzle out of it, in which each piece was one of the squares.

"It was at this stage that Brooks's mother made the key discovery of the whole investigation. She tackled Brooks's puzzle and eventually succeeded in putting the pieces together to form a rectangle. But it was not the squared rectangle which Brooks had cut up! The Important Members met in emergency session.

"We had sometimes wondered whether it was possible for two different perfect rectangles to have the same shape. We would have liked to obtain two such rectangles with no common reduced element, and thus to get a perfect square by the construction shown in Figure 7. The hatched regions in this diagram represent the two perfect rectangles. Two unequal squares are added to make the large perfect square. But no rectangles of the same shape had hitherto appeared in our catalog, and we had reluctantly come to believe that the phenomenon was impossible. Mrs. Brooks's discovery renewed our hopes, despite the fact that her rectangles failed in the worst possible way to have no common reduced element.

"There was much excited discussion at the emergency session. Eventually the Important Members calmed down sufficiently to draw the Smith diagrams of the two rectangles. Inspection of these soon made clear the relationship between them.

"The second rectangle is shown in Figure 8, and its Smith diagram in Figure 9. It is evident that the network of Figure 9 can be obtained from that of Figure 5 by placing the terminals P and P' at the same point. As P and P' happen to have the same electrical potential in Figure 5, this operation causes no change in the currents in the individual wires, no change in the total current, and no change in the potential difference between the poles. We thus have

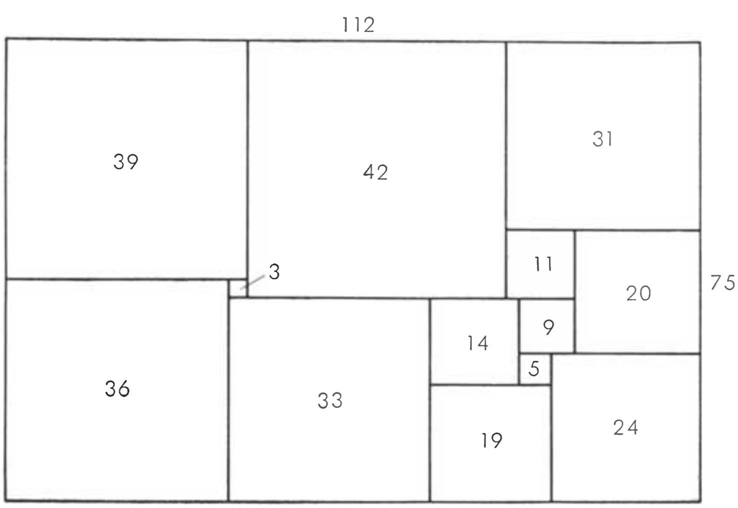

Figure 8

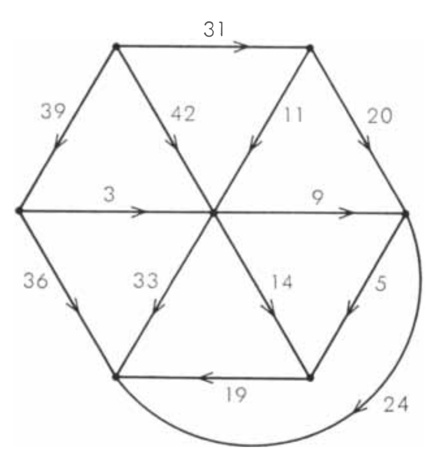

Figure 9

a simple electrical explanation for the fact that the two rectangles have the same reduced sides and elements.

"But why do P and P' have the same potential in Figure 5? Before the emergency session had broken up it had also obtained an answer to this question. The explanation depends on the fact that the network can be decomposed into three parts meeting only at poles A_1 and A_2 and the terminal A_3. One of these parts consists solely of the wire joining A_2 and A_3. A second part is made up of the three wires meeting at P', and a third is constituted by the remaining nine wires. Now the third part has threefold rotational symmetry, with P as the center of rotation. Moreover, current enters or leaves this part of the network only at A_1, A_2 and A_3, which are equivalent under the symmetry. This is enough to insure that if any potentials whatever are applied to A_1, A_2 and A_3, the potential of P will be their average. The same argument applied to the second part of the network shows that the potential of P' must also be the average of the potentials of A_1, A_2 and A_3. Hence P and P' have the same potential, whatever potentials are applied to A_1, A_2 and A_3; and in particular they have the same potential when A_1 and A_2 are taken as poles in the complete network and the potential of A_3 is fixed by Kirchhoff's laws.

"The next advance was accidentally made by the present writer. We had just seen Mrs. Brooks's discovery completely explained in terms of a simple property of symmetrical networks. It seemed to me that it should be possible to use this property to construct other examples of pairs of perfect rectangles with the same reduced elements. The obvious thing to do was to replace the third part of the network in Figure 5 by another network having threefold rotational symmetry about a central terminal. But this could be done only under severe limitations which should now be explained.

"It can be shown that the Smith diagram of a squared rectangle is always planar, that is, it can be drawn in a plane with no crossing wires. Moreover, the drawing can always be made so that no circuit separates the two poles. There is also a converse theorem which states that if an electrical network of unit resistances can be drawn in a plane in this way, then it is the Smith diagram of some squared rectangle.

"My problem was to replace the third part of the network shown in Figure 5 by a new symmetrical network with center P, in such a way that the complete network would not only be planar, but remain planar when P and P' were coalesced. After a few trials I found two closely related networks satisfying these conditions (*Figures 10 and 11*). As was expected, each diagram allowed of the coalescence of P and P', and so gave rise to two squared rectangles with the same reduced elements. But all four rectangles had the same reduced sides, and this result was quite unexpected.

"Essentially the new discovery was that the rectangles corresponding to Figures 10 and 11 have the same shape, although their reduced elements are not all the same. A simple theoretical explanation of this was soon found. The two networks have the same structure, apart from the choice of poles, and therefore the rectangles have the same full horizontal side. Moreover, the networks remain identical when poles are coalesced, and therefore the two rectangles have the same vertical side. We felt, however, that this explanation did not probe deep enough, since it made no reference to rotational symmetry.

"We agreed to refer to the new phenomenon as 'rotor-stator' equivalence. It was always associated with a network which could be decomposed into two parts, the 'rotor' and the 'stator,' with

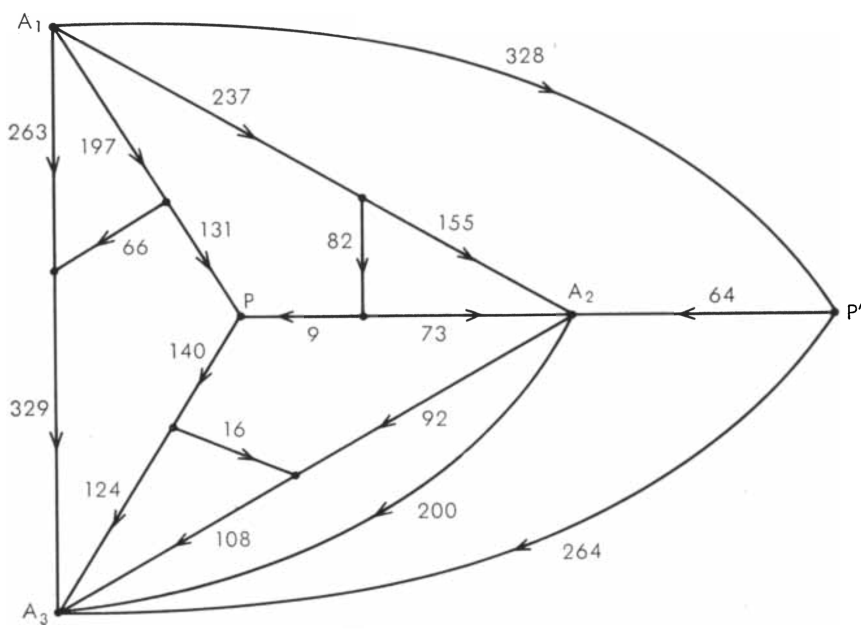

Figure 10

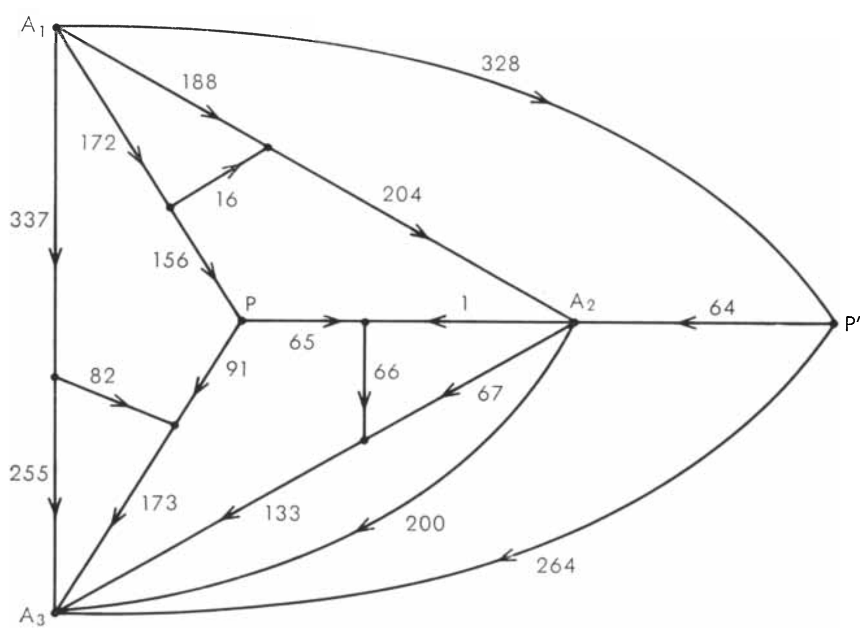

Figure 11

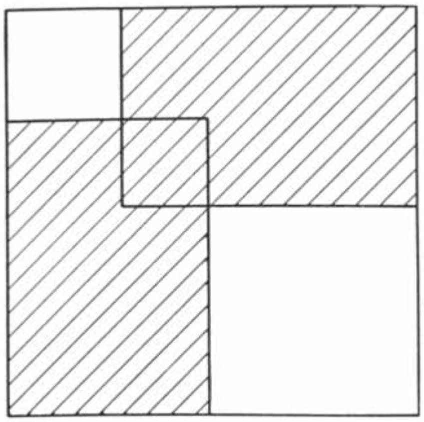

Figure 12

the following properties. The rotor had rotational symmetry, the terminals common to the rotor and stator were all equivalent under the symmetry of the rotor, and the poles were terminals of the stator. In Figure 10, for example, the stator is made up of the three wires joining P' to A_1, A_2 and A_3, and the wire linking A_2 with A_3. A second network could then be obtained by an operation called 'reversing' the rotor. With a properly drawn figure this could be explained as a reflection of the rotor in a straight line passing through its center. Thus starting with Figure 10 we can reflect the rotor in the line PA_3, and so obtain the network of Figure 11.

"After studying a few examples of rotor-stator equivalence the researchers convinced themselves that reversing the rotor made no difference to the full sides of the rectangle, and no difference to the currents in the wires of the stator. But the currents in the rotor might change. We were able to obtain satisfactory proofs of these results only at a much later stage.

"A tantalizing question now arose. What was the least possible number of common elements in a rotor-stator pair of perfect rectangles? Those of Figures 10 and 11 had seven common elements, three of which corresponded to currents in the rotor. The same rotor with a stator consisting of a single wire A_2A_3 and the extra terminal A_1 gave two perfect rectangles of the 16th order with four common elements. Using a one-wire stator there seemed no theoretical reason why we should not obtain a pair of perfect rectangles having only one element, corresponding to the stator, in common. But we saw that if we could do this we could also obtain a perfect square. For with the rotors of threefold symmetry which we were studying, a one-wire stator always respresented a corner element of each corresponding rectangle. From two perfect rectangles with only a corner element in common we could expect to obtain a perfect square by the construction illustrated in Figure 12. Here the hatched regions represent the two rectangles. The square in which they overlap is the common corner element.

"Naturally we got to work at once calculating rotor-stator pairs. So it came to pass that Smith and Stone sat down one day to compute one complicated pair while Brooks, unknown to them, worked on another in a different part of the College. After some hours Smith and Stone burst into Brooks's room crying, 'We have a perfect square!', to which Brooks replied, 'So have I.' Both squares were of the 69th order. But Brooks went on to experiment with simpler rotors, and so obtained a perfect square of the 39th order.

"There are three more episodes worth mentioning in the history of the perfect square, though each may seem like an anticlimax. To begin with we kept adding to the list of simple perfect rectangles of the 13th order. One day we found that two of these rectangles had the same shape and no common element. They gave rise to a perfect square of the 28th order by the construction of Figure 7. Later we found a 13th-order perfect rectangle which could be combined with one of the 12th order and one extra component square to give a perfect square of the 26th order. If the merit of a perfect square is measured by the smallness of its order, then the empirical method of cataloging perfect rectangles had proved superior to our beautiful theoretical method.

"Other researchers have used the empirical method with spectacular results. R. Sprague of Berlin fitted a number of perfect rectangles together in a most ingenious way to produce a perfect square of the 55th order. This was the first perfect square to be published (1939). More recently T. H. Willcocks of Bristol obtained a perfect square of the 24th order (*Figure 13*). This still holds the low-order record.

"In case any reader would like to work on perfect rectangles, here are two unsolved problems. What is the smallest possible order for a perfect square? Is there a simple perfect rectangle twice as long as it is high? Simple perfect squares have already been obtained by extensions of the theoretical method."

Figure 13

Last month's figure-tracing puzzle can be solved with as few as 13 corner turns. Start at the second node from the left on the large triangle. Move up and to the right as far as possible, then left, then down and right to the base of the triangle, up and right, left as far as possible, down and right, right to the corner of the large triangle, up to the top of the triangle, down to the triangle's left corner, all the way around the circle, right to the third node on the triangle's base, up and left as far as possible, right as far as possible, then down and left to the base.

The cord-and-ring puzzle is solved as follows. Loosen the center loop enough so that the ring can be pushed up through it. Hold the ring against the front side of the panel while you seize the double cord where it emerges from the center hole. Pull the double cord toward you. This will drag a double loop out of the central hole. Pass the ring through this double loop. Now reach behind the panel and pull the double loop back through the hole so that the cord is restored to starting position. It only remains to slide the ring down through the center loop and the puzzle is solved.

Many readers have pointed out an error in the answer in September to problem No. 2. Three airplanes are quite sufficient to insure the flight of one plane around the world. There are many ways this can be done, but the following seems to be the most efficient. It uses only five tanks of fuel, allows the pilots of two planes sufficient time for a cup of coffee and a sandwich before refueling at the base, and there is a pleasing symmetry in the procedure.

Planes A, B and C take off together. After going 1/8 of the distance, C transfers 1/4 tank to A and 1/4 to B. This leaves C with 1/4 tank; just enough to get back home.

Planes A and B continue another 1/8 of the way, then B transfers 1/4 tank to A. B now has 1/2 tank left, which is sufficient to get him back to the base where he arrives with an empty tank.

Plane A, with a full tank, continues until it runs out of fuel 1/4 of the way from the base. It is met by C which has been refueled at the base. C transfers 1/4 tank to A, and both planes head for home.

The two planes run out of fuel 1/8 of the way from the base, where they are met by refueled plane B. Plane B transfers 1/4 tank to each of the other two planes. The three planes now have just enough fuel to reach the base with empty tanks.